SIDDHARTH INSTITUTE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)

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Accredited by NAAC with 'A+' Grade)

Puttur -517583,Tirupathi District, A.P. (India)

DEPARTMENT OF

CSE (ARTIFICIAL INTELLIGENCE)

Academic Year 2025-26

Project Viva-Voce Examinations Date : 10.04.2026

Title of Project

Towards Green and Trustworthy 6G:Federated Learning Approaches for Robust Privacy.

Batch Details

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OBJECTIVE

1. This project aims to build a privacy-focused machine learning system for future 6G networks.
2. Instead of collecting all data in one place, it uses Federated Learning (FL), where multiple devices train a model together without sharing their raw data. This helps in:
   > Protecting user privacy
   > Reducing network traffic
3. A Flask-based web application is developed where users can:
   > Register and log in
   > Upload beamforming datasets
   > Choose machine learning algorithms
   > Get real-time predictions
4. The system works by simulating FL training rounds, where model updates from different devices are securely combined (averaged) to improve performance.
5. Overall, the project shows how Federated Learning can efficiently support intelligent and secure systems in 6G networks.

ABSTRACT

1. 6G technology will connect billions of smart devices and provide very fast communication (low latency) for advanced applications like autonomous vehicles and holograms.

2. Traditional AI systems rely on sending data to the cloud, which:
   > Overloads the network
   > Risks user privacy

3. To solve this, this project uses Federated Learning (FL), where:
   Devices train models locally
   Only encrypted updates are shared with a central server

4. A Flask-based web application is developed where users can:
   > Register and log in
   > Upload beamforming datasets
   > Select algorithms like Random Forest, CNN, MLP, etc.
   > Get real-time prediction results

LITERATURE REVIEW

Title of the Paper	Authors	Year	Implications
Blockchain empowered Federated Learning for 6G.	Xu et al.	2023	Combine FL with blockchain for traceability and trust in 6G; emphasizes audit trails and secure updates.
Federated Learning for Edge Intelligence in 6G Networks.	Yang et al.	2022	Overview of enabling FL in 6G, covers latency, non-IID issues, and model convergence.
Federated Learning in 6G: A Survey on Enabling Technologies.	Luong et al.	2021	In-depth survey on FL challenges and architectures in 6G ecosystems.

Title of the Paper	Authors	Year	Implications
Adaptive Federated Learning in Resource Constrained Edge Systems.	Wang et al.	2021	Proposes adaptive scheduling strategies to handle client heterogeneity and stragglers
Practical Secure Aggregation for Privacy-Preserving ML.	Bonawitz et al.	2021	Introduces secure aggregation protocols enabling privacy-preserving global model updates.
Federated Learning for Edge Intelligence: Applications and Challenges	Zhang et al.	2021	Highlights edge-based FL deployment, communication savings, and scalability.

EXISTING SYSTEM

Limitations of Traditional AI Systems (6G)

•Uses centralized cloud-based training

•Devices send raw data to central servers continuously

Major Problems

•**Network congestion:** Massive real-time data overloads backhaul links

•**High latency:** Delays make it unsuitable for real-time applications (autonomous driving, AR/VR)

•**Privacy risks**: Sensitive user data is exposed during transmission and storage

•**Not scalable:** Cannot efficiently support billions of 6G edge devices

•**No personalization:** Models are generic and not adapted to local device data

•**High energy consumption:** Continuous data transmission drains device and network resources

•**Security vulnerabilities:** Prone to attacks like data leakage, model poisoning, and reconstruction attacks

•**Single point of failure:** Central server failure affects the entire system

•Traditional AI systems are inefficient, insecure, and not suitable for scalable 6G environments, highlighting the need for decentralized approaches like Federated Learning.

PROPOSED SYSTEM

Overview

- Uses Federated Learning (FL) for 6G networks
- Devices train on local private data
- Share only encrypted updates (no raw data)
- Server combines updates → Global model

Application (Flask Web App)

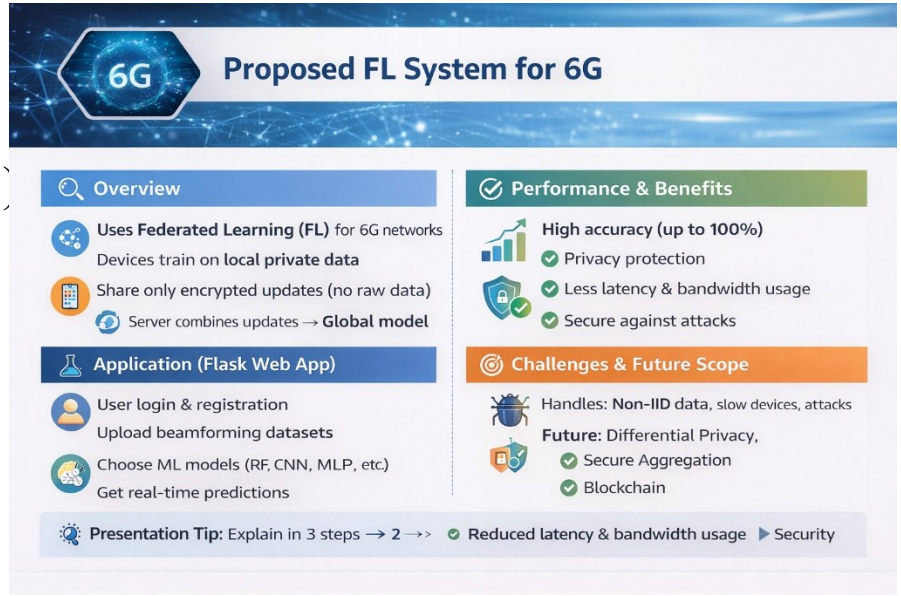
- User login & registration
- Upload beamforming datasets
- Choose ML models (RF, CNN, MLP, etc.)
- Get real-time predictions

Performance & Benefits

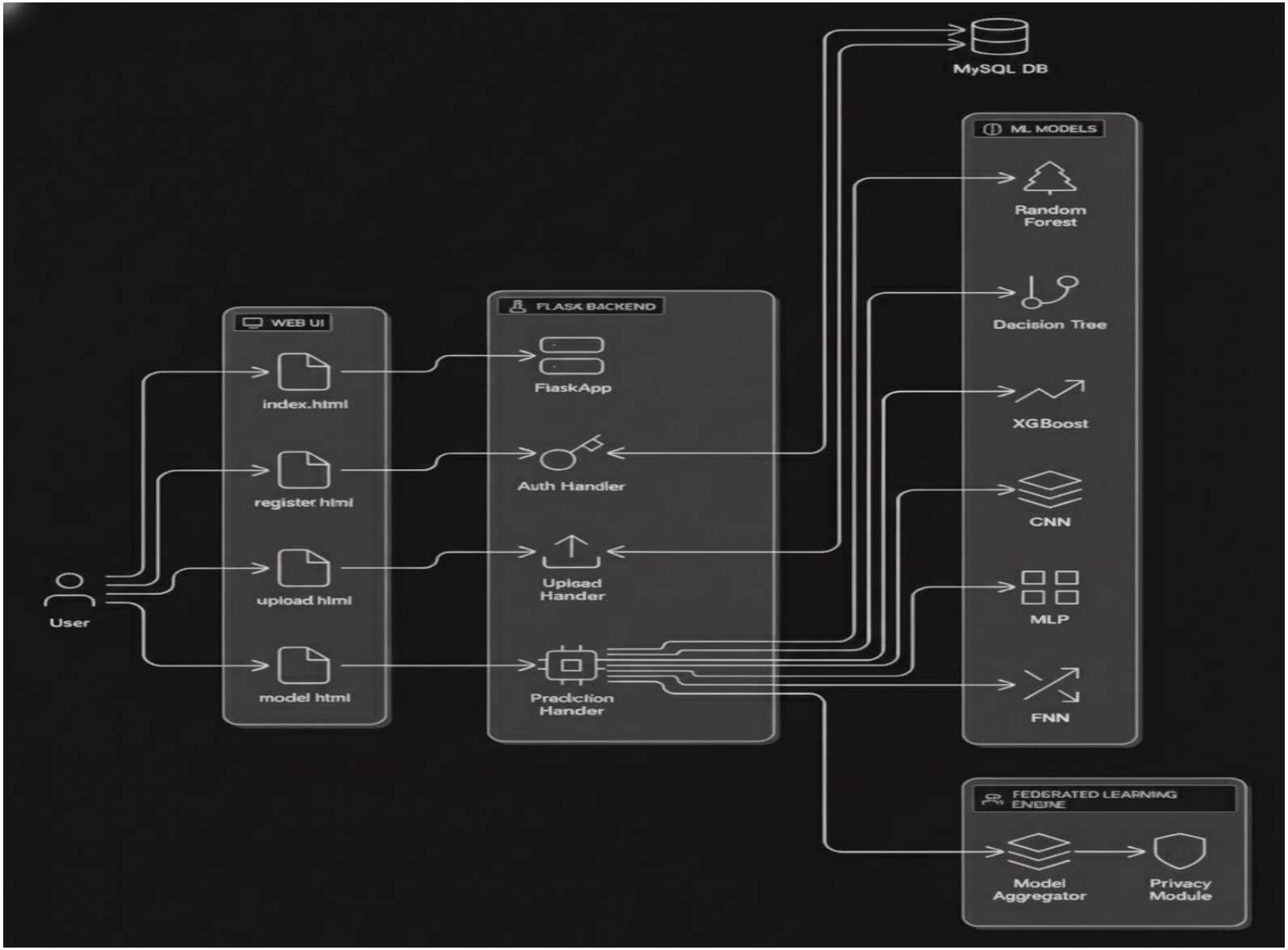
- High accuracy (up to 100%)
- Privacy protection
- Less latency & bandwidth usage
- Secure against attacks

Challenges & Future Scope

- Handles: Non-IID data, slow devices, attacks
- Future: Differential Privacy, Secure Aggregation, Blockchain

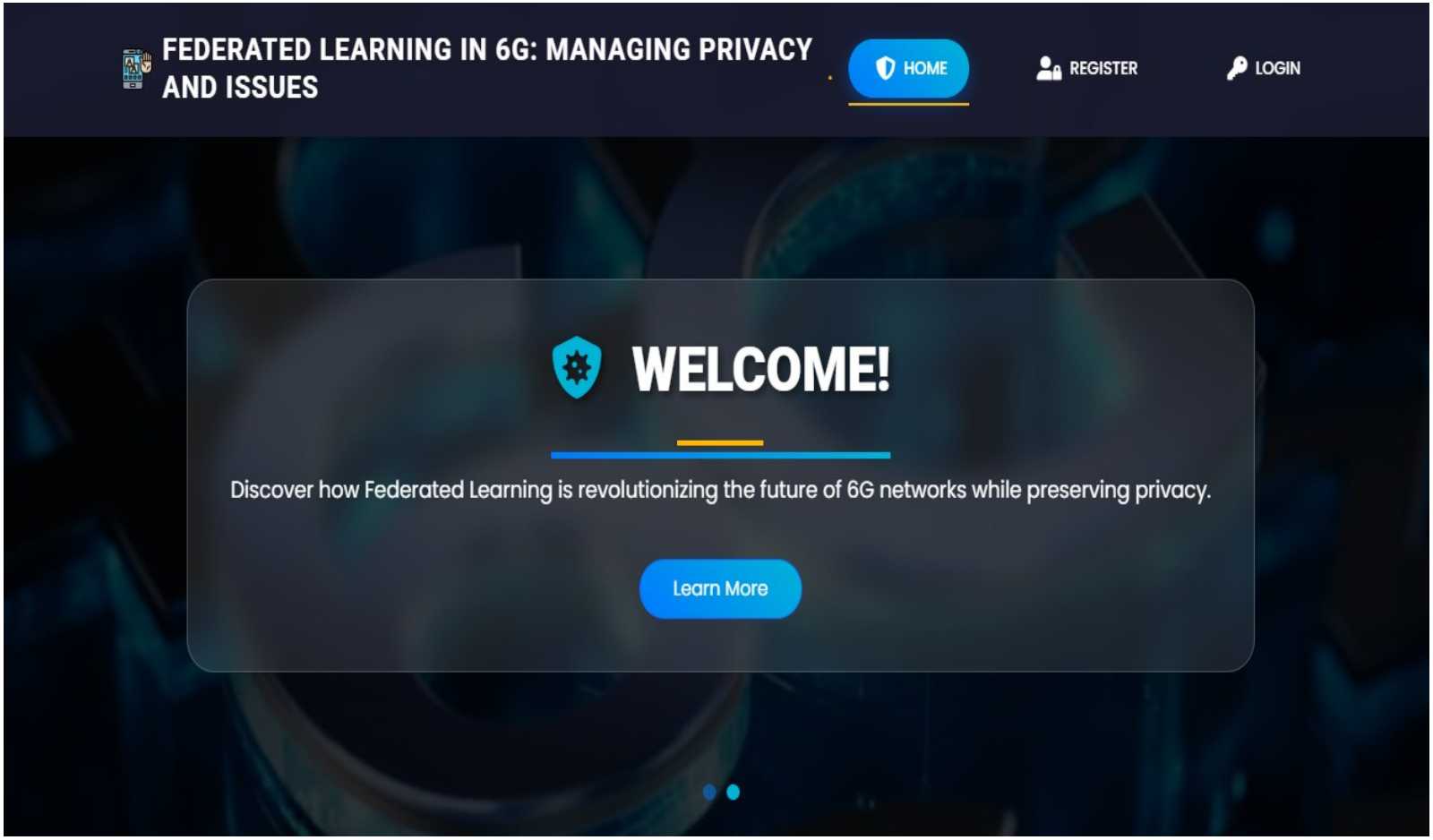


ARCHITECTURE DIAGRAM

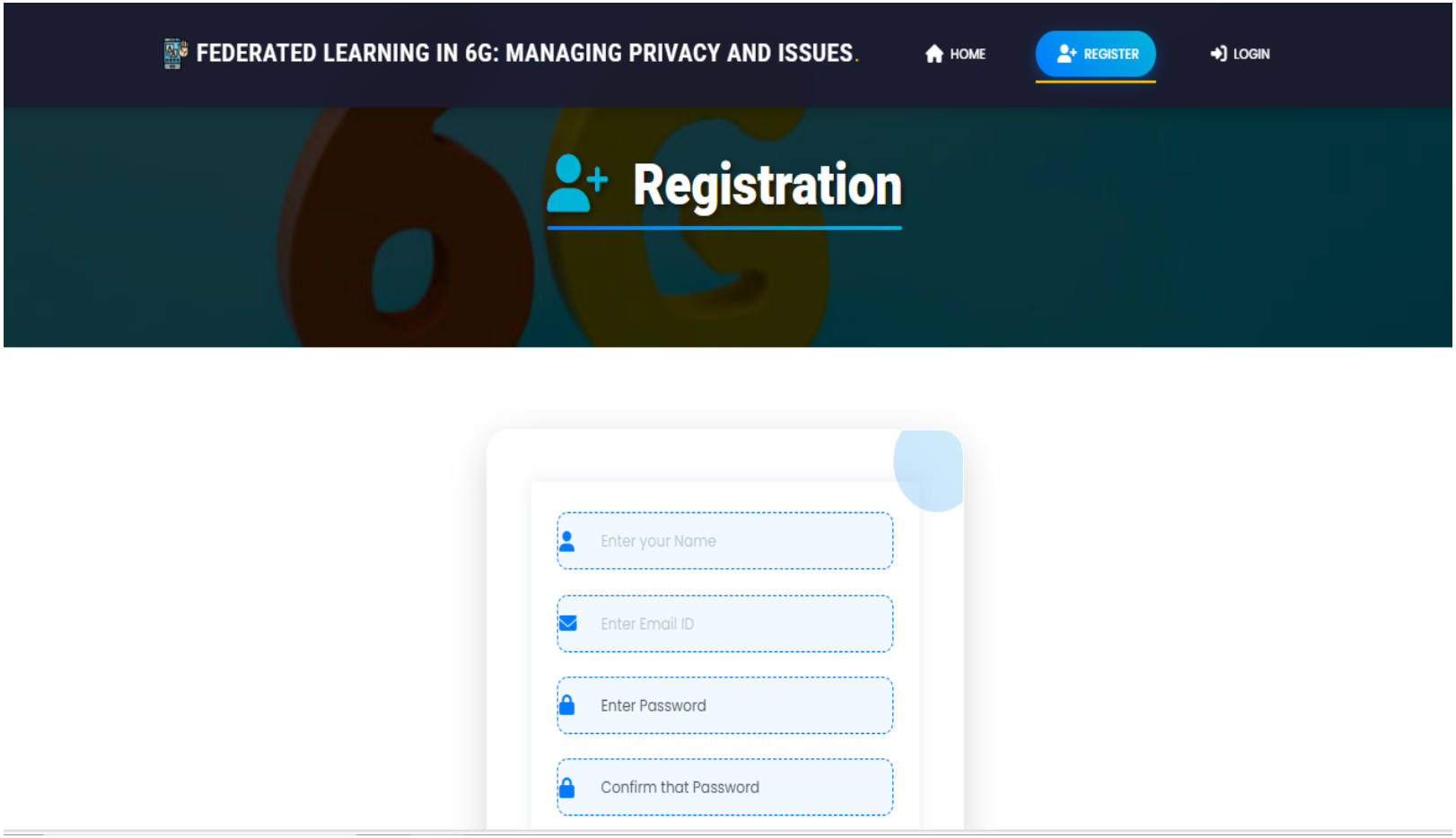


MODULES

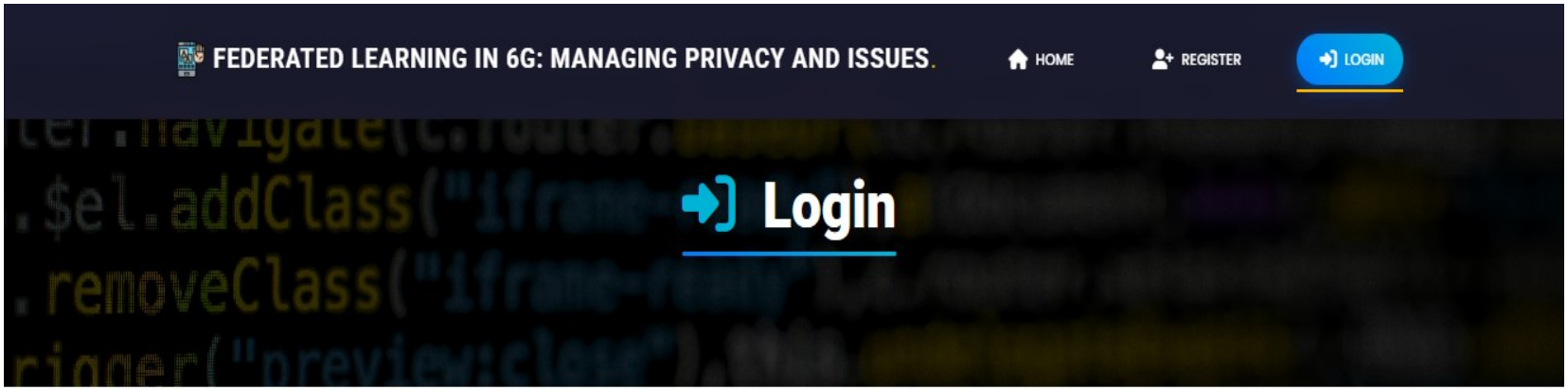
HOME PAGE: The Home Page serves as the landing page of your application.
It provides an overview of the project's features, objectives, and benefits. Users can navigate to other sections of the application from this page.

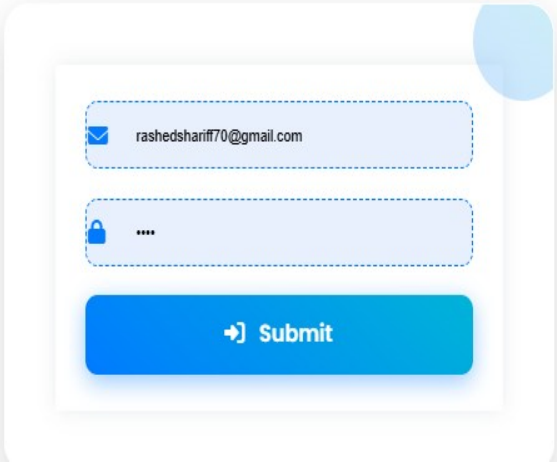


REGISTRATION PAGE: The Registration Page allows new users to create an account with the application. It typically includes fields for entering personal information such as name, email,password, and possibly other details like phone number or address. Users need to fill out this form to gain access to the application's features.

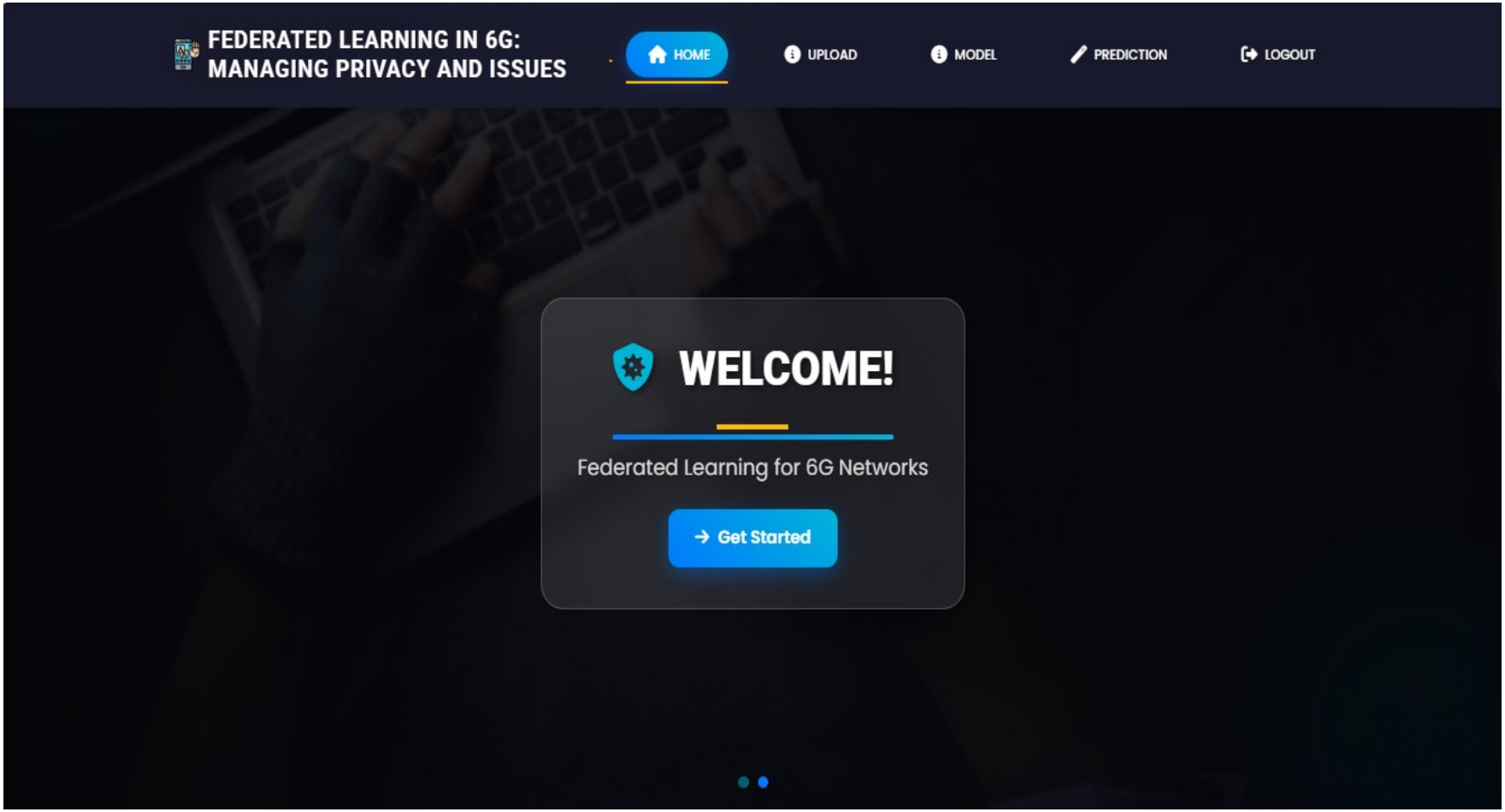


LOGIN: The login page allows registered users to securely access the application. User have to provide register email and password.

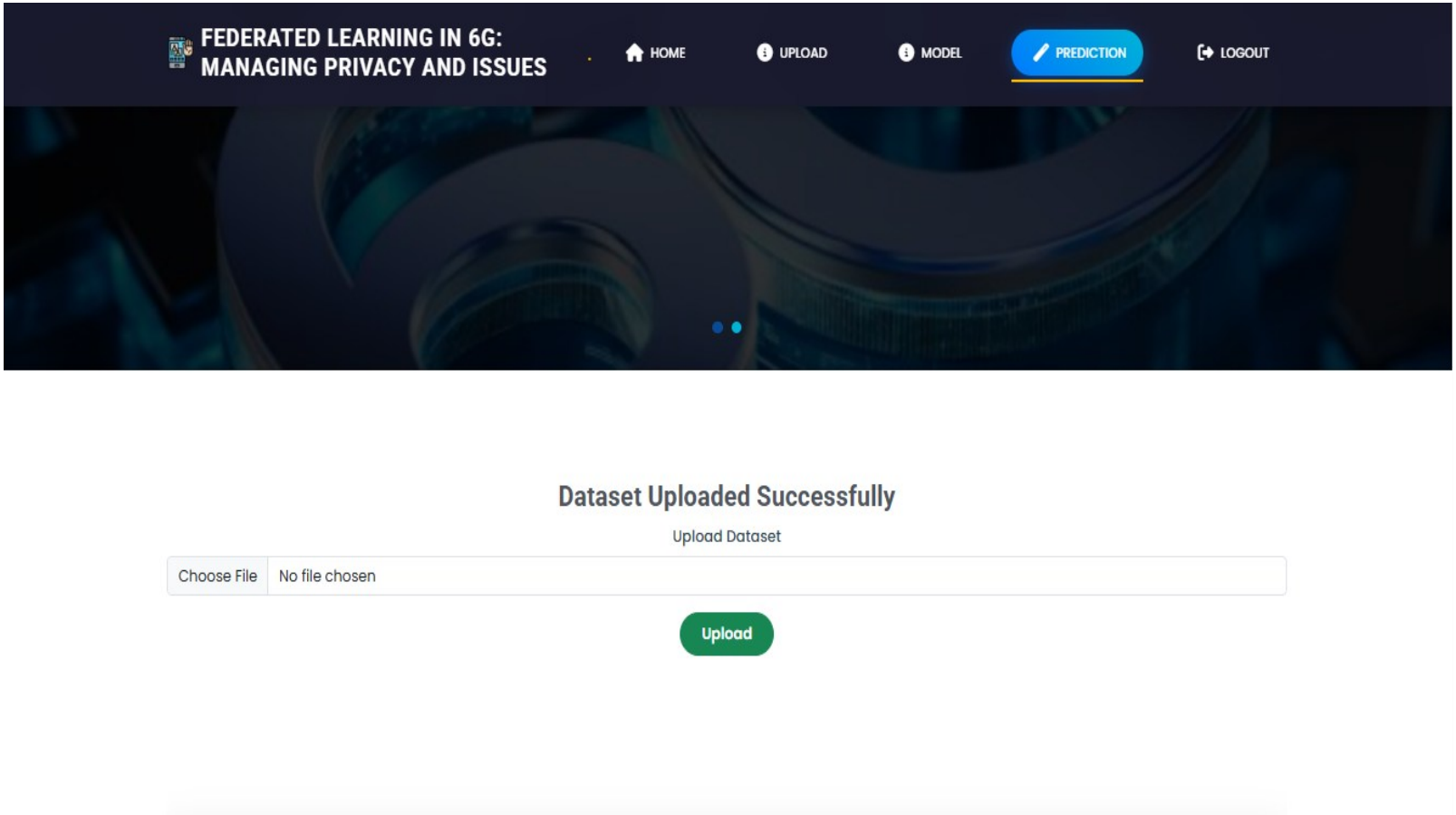




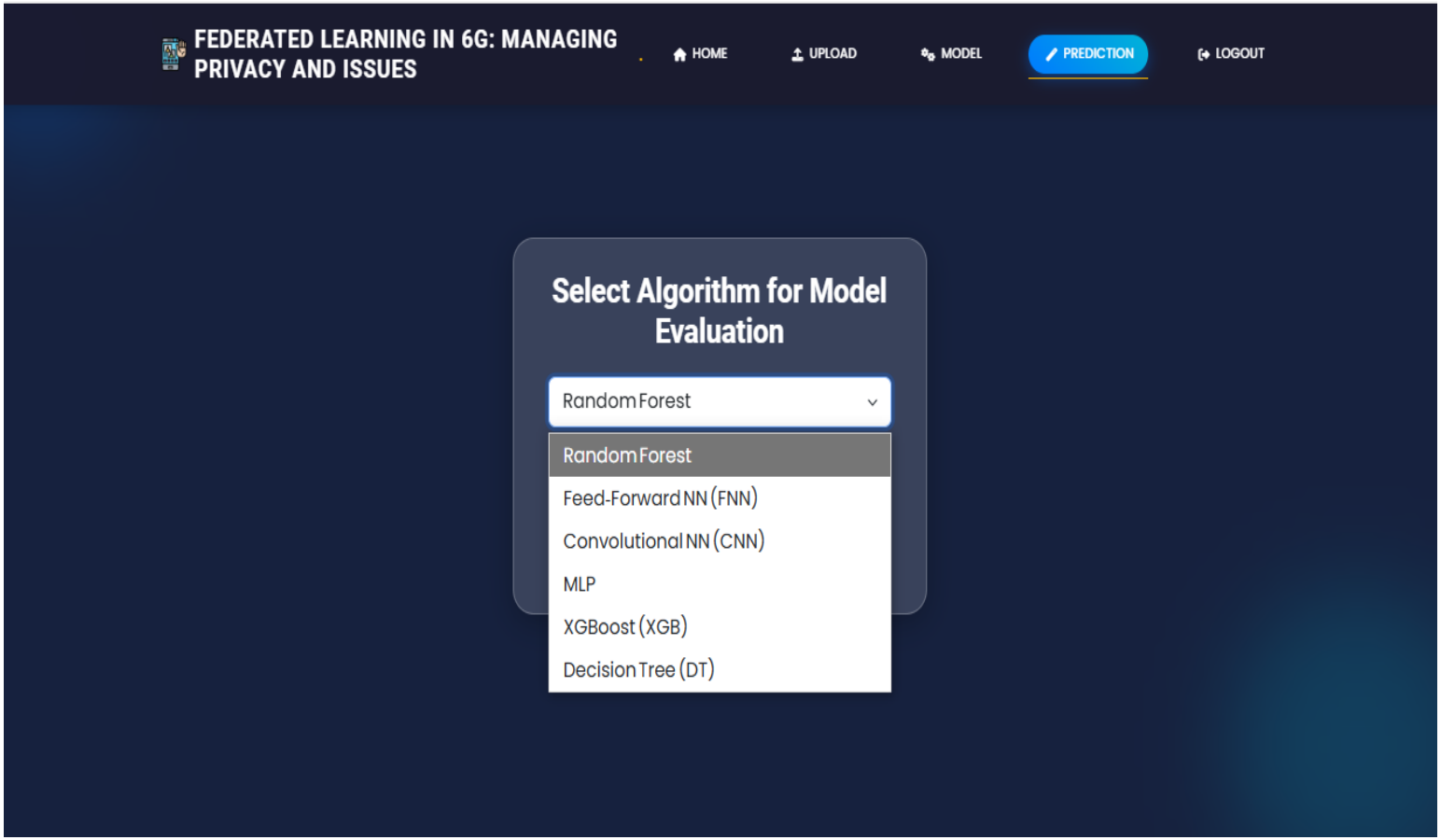
HOME PAGE AFTER LOGIN



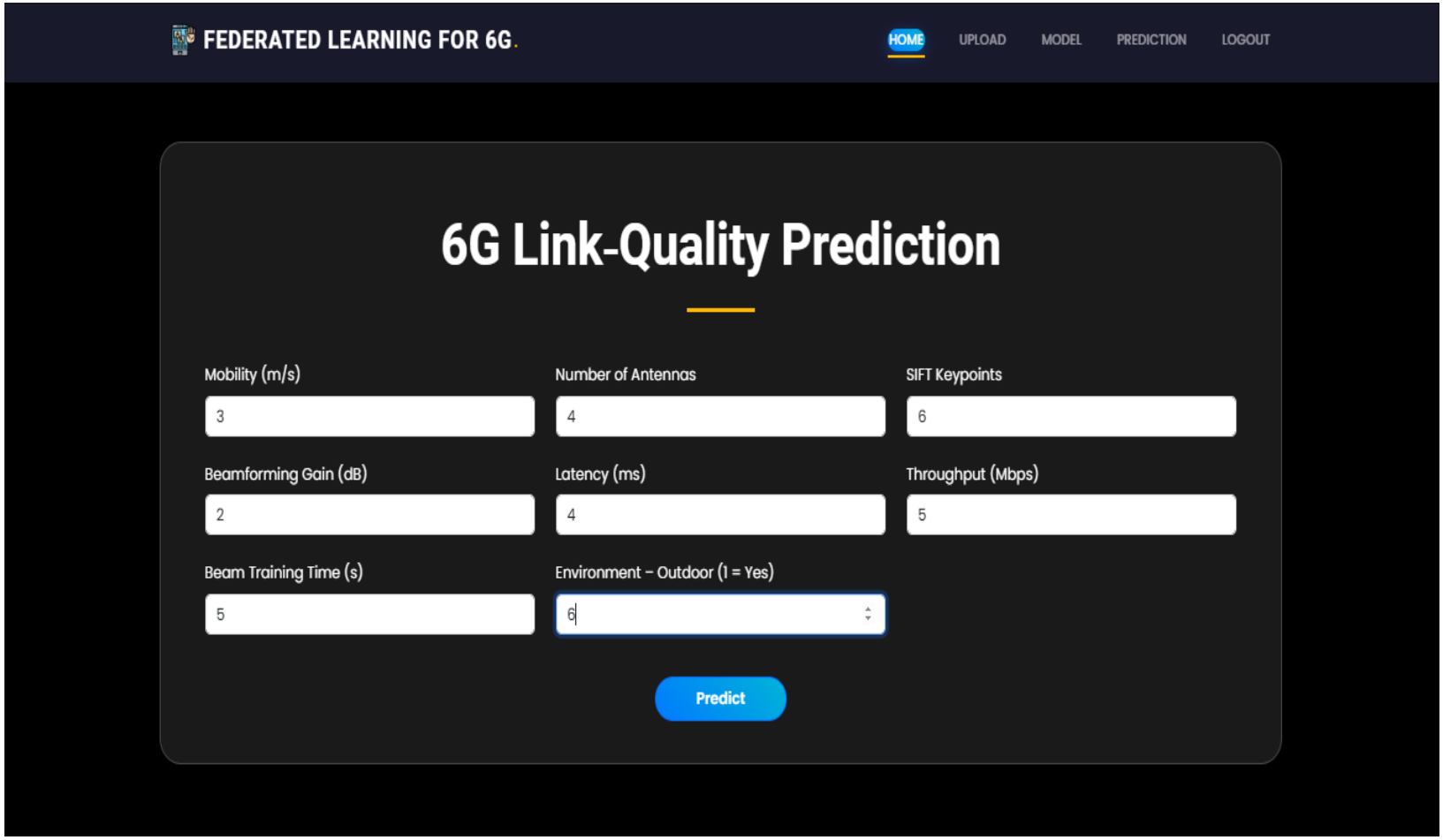
UPLOAD PAGE: Here you can upload the dataset and view the dataset



MODEL SELECTION: Here user have to select the model and email, password view the accuracy so that we know which algorithm is giving the highest accuracy.



PREDICTION PAGE: This is the prediction page here user have to provide values of a 20 features and our model is predict the final output.



RESULT EVALUATION

PREDICTION OUTCOME

•Models classified beamforming results as Success or Failure

•Evaluation performed under simulated federated learning rounds

•No raw data shared; only model parameters aggregated

Performance Results

•Decision Tree, Random Forest, XGBoost

→ 100% accuracy in predicting Success/Failure

•FNN, CNN, MLP

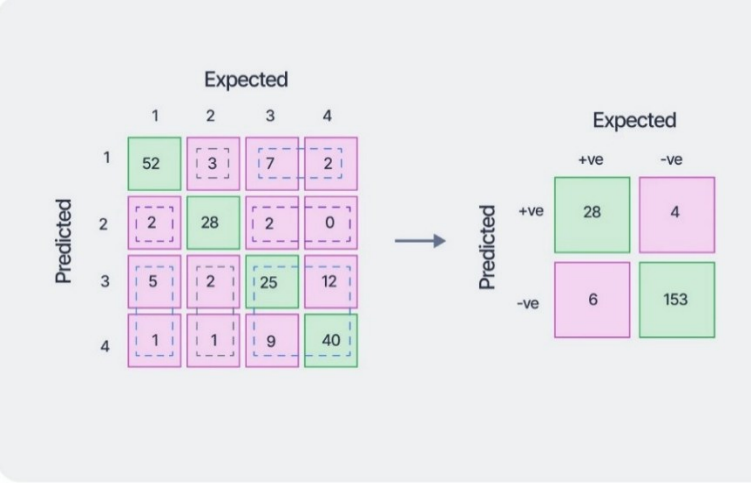
→ High accuracy with minor misclassification

Confusion Matrix Insight

•**Tree-based models showed:**

- Zero false positives
- Zero false negatives

•Indicates perfect beamforming decision reliability



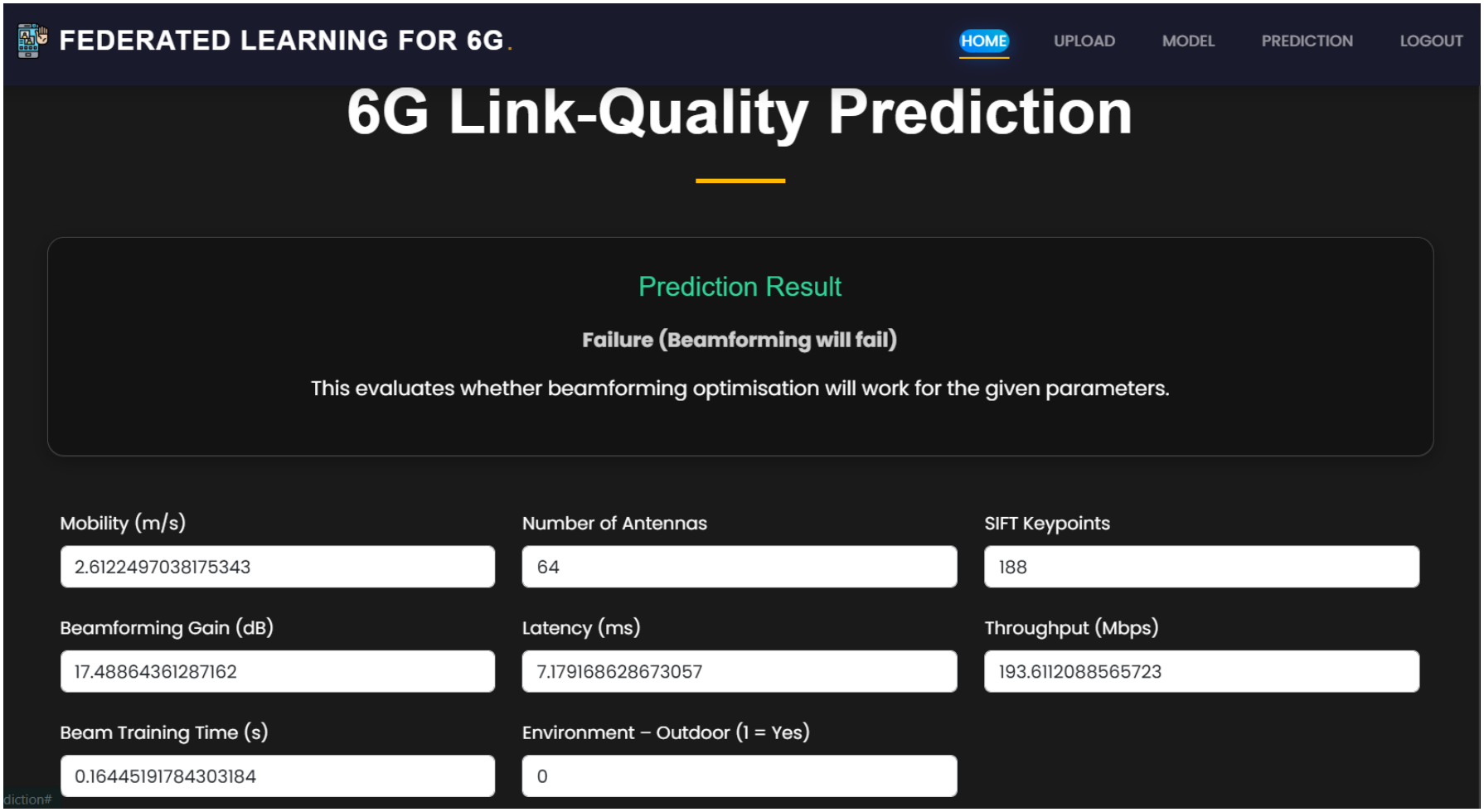
Federated Learning Impact

•✓ Preserved data privacy

•✓ Reduced communication overhead

•✓ Maintained centralized-model-level accuracy

RESULT PAGE :After prediction show whether the beamforming will success or failure.



CONCLUSION

- This project proposes a privacy-preserving federated learning framework to predict 6G beamforming success or failure. Traditional cloud-based ML systems face latency, scalability, and privacy issues, making them unsuitable for future 6G applications like autonomous mobility and holographic communication.
- Our system uses Federated Learning, where edge devices train models locally and share only model parameters, ensuring data privacy. The framework is implemented using Python Flask with features such as secure authentication, dataset upload, model selection, and real-time prediction.
- Six ML models are supported: Random Forest, XGBoost, FNN, CNN, MLP, and Decision Tree.

  Tree-based models achieved 100% accuracy in simulations, proving their effectiveness for beamforming classification

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McMahan, B., Moore, E., Ramage, D., Hampson, S., & y Arcas, B. A. (2017). *Communication-Efficient Learning of Deep Networks from Decentralized Data*. Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS), PMLR. https://arxiv.org/abs/1602.05629

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